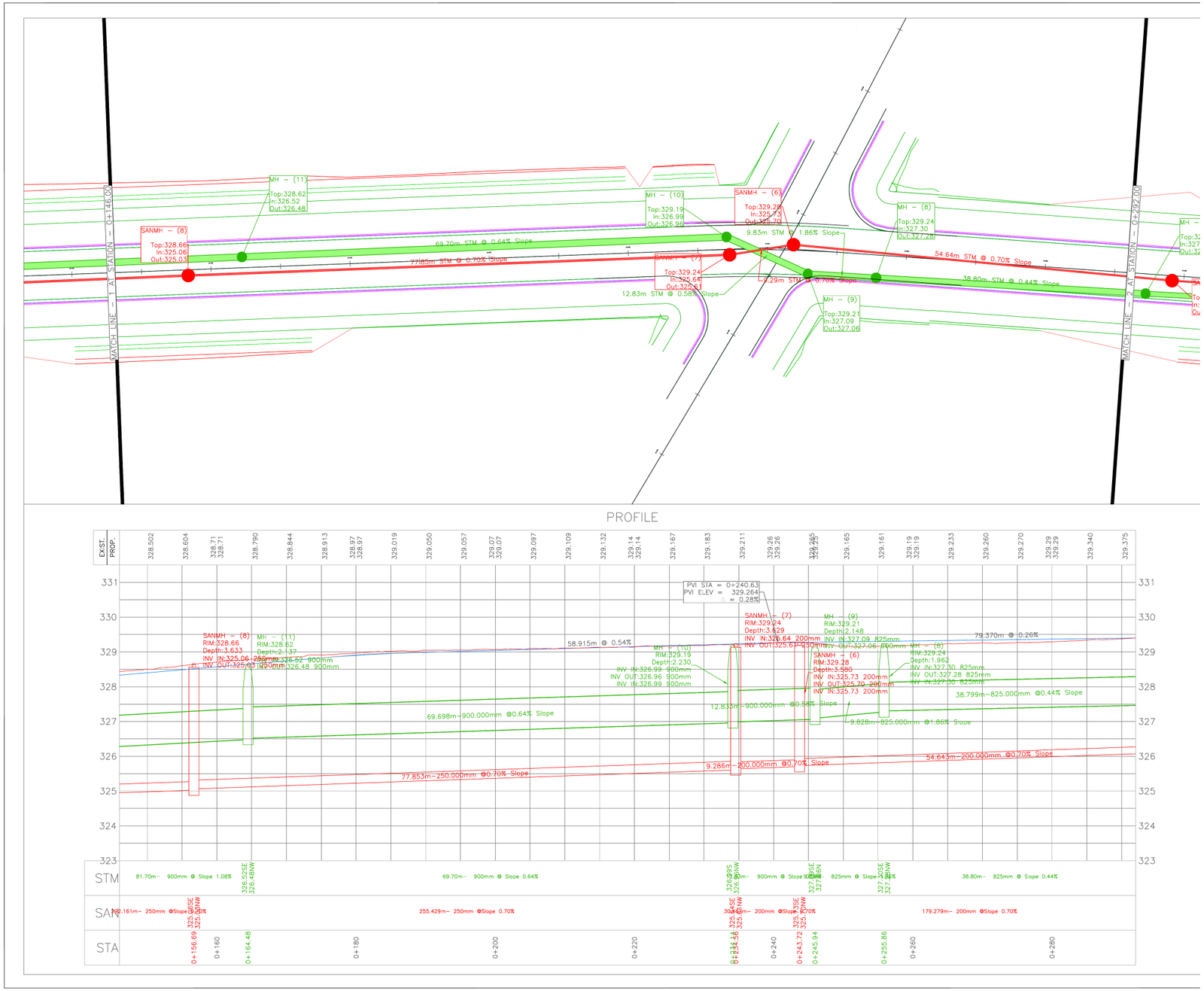


CIVIL 3D DESIGN PORTFOLIO

Municipal Road and Sewer Design

Conceptual plan and profile drawing package

SELECTED CAPABILITIES

- **Road corridor and intersection coordination**
- **Storm and sanitary plan/profile production**
- **Conceptual GIS/LiDAR existing-condition modelling**

4 PLAN/PROFILE + 3 SECTION SHEETS

ROAD / STORM / SANITARY / CROSS-SECTIONS

CONCEPTUAL - NOT FOR CONSTRUCTION

Coordinated municipal road and sewer design

Design intent translated into a controlled Civil 3D model

This independent portfolio uses public GIS and LiDAR-derived information to demonstrate a complete conceptual workflow: existing-ground development, road and intersection modelling, storm and sanitary coordination, and plan/profile sheet production.

PROJECT TYPE

Municipal road reconstruction and underground servicing

DESIGN PLATFORM

Autodesk Civil 3D with data shortcuts, XREFs and sheet sets

PRIMARY OUTPUT

Four plan/profile and three road cross-section sheets

PROJECT STATUS

Conceptual portfolio project - not for construction

DELIVERED IN THE MODEL AND DRAWING PACKAGE

01 Existing context

Terrain and mapping inputs organized into an existing-ground reference.

02 Road system

Alignment, profile, corridor, intersection and lane-width transition coordination.

03 Buried services

Storm and sanitary networks displayed in plan and profile with structures and bands.

04 Sheet production

View frames, match lines, XREF coordination, annotation and colour publication.

PROFESSIONAL BOUNDARY

The work demonstrates modelling and drawing-production skills. Existing conditions and buried utilities require survey, records review and field verification. The package is not approved, sealed, tendered or construction-ready.

Criteria applied to the conceptual design

Defined controls before detailed modelling

The following criteria were used as portfolio design controls based on applicable local municipal infrastructure requirements. They describe the modelled intent; they do not replace project-specific standards, survey, calculations, approvals or licensed engineering review.

DESIGN CONTROL	MODELLED INTENT	CURRENT EVIDENCE	NEXT QA GATE
Road grades	Geometric controls and vertical transitions	FG profiles and corridor references present	Check criteria and tie-in elevations
Lane drainage	Typical 2.0% crossfall to gutter	Road and offset profile workflow demonstrated	Confirm every region and superelevation case
Lane transition	3.5 m lane transitioning to 3.0 m	Corridor width transition modelled	Review taper geometry and feature-line continuity
Storm cover	Target 1.5 m minimum	Target documented; exported minima include values below target	Rebuild network, resolve exceptions and verify report
Sanitary cover	Target 2.5 m minimum	Intended criterion; independent cover export not available	Export network report and verify every pipe
Pipe geometry	Positive slope, coordinated inverts and structures	Plan/profile networks and bands demonstrated	Check cover, clearances, crossings and connectivity

SENIOR REVIEW POSITION

The cover values are design targets, not completed compliance claims. The current storm export contains exceptions below the target, and an equivalent sanitary verification export was not available. Before any real use, rebuild the networks, resolve exceptions, confirm downstream capacity and hydraulics, and obtain survey and licensed engineering review.

Controlled model-to-sheet production

One source of truth, staged through explicit QA gates

The production approach separates source data, design objects and presentation files so revisions can be traced and sheets can be rebuilt without manual drafting drift.

01	SOURCE DATA **Public GIS and LiDAR inputs**	QA GATE	Check coordinate system, extents, units and data limitations.
02	BASE MODEL **EG surface and reference geometry**	QA GATE	Review surface boundaries, profiles and anomalous elevations.
03	ROAD DESIGN **Alignment, FG profile, corridor and intersection**	QA GATE	Check tie-ins, lane widths, crossfall, targets and regions.
04	PIPE NETWORKS **Storm and sanitary pipes and structures**	QA GATE	Check connectivity, slopes, inverts, cover and crossings.
05	PRODUCTION **XREFs, view frames, match lines and profile bands**	QA GATE	Check ownership, styles, station ranges and annotation.
06	PUBLICATION **Sheet set, colour PDF and portfolio wrapper**	QA GATE	Check plotting, legibility, sheet continuity and final PDF.

REVISION PRINCIPLE Revise the owning Civil 3D object or source drawing, synchronize references, rebuild the corridor and networks, then regenerate and visually inspect the sheets.

Ownership, authorization and closeout workflow

Authorization is established before detailed design

A real project begins by confirming ownership, regulatory scope and the approval pathway. CLI-ECA screening applies to municipal sanitary collection and stormwater works; it does not replace planning, environmental assessment or other permits.

01

Confirm scope and ownership

Map existing/proposed assets. Confirm who owns and operates the works, whether new works will be transferred to the municipal owner, and whether the work is repair, maintenance, a lateral connection or an alteration.

02

Screen the approval pathway

Obtain the owner's current CLI-ECA and project requirements. Check eligibility, prescribed-person status, receiving-system constraints, affected components and required pre-consultation.

03

Schedule D or Schedule C

If every condition for a future alteration in Schedule D is satisfied, follow the pre-authorized route and complete the applicable records. If not, the owner or authorized party must obtain a Schedule C amendment before alteration.

04

Design-stage submission

Prepare the design brief, drawings, design sheets, calculations, criteria-exception list and applicable records such as SS1/SS2 for sewage collection works or SW1/SW2/SW3 for stormwater works. Obtain required owner and licensed-practitioner verification.

05

Detailed design and QA/QC

Coordinate survey, utility records, geotechnical information, hydrology/hydraulics, road geometry, pipe capacity, cover, crossings, constructability, quantities, specifications and independent checking.

06

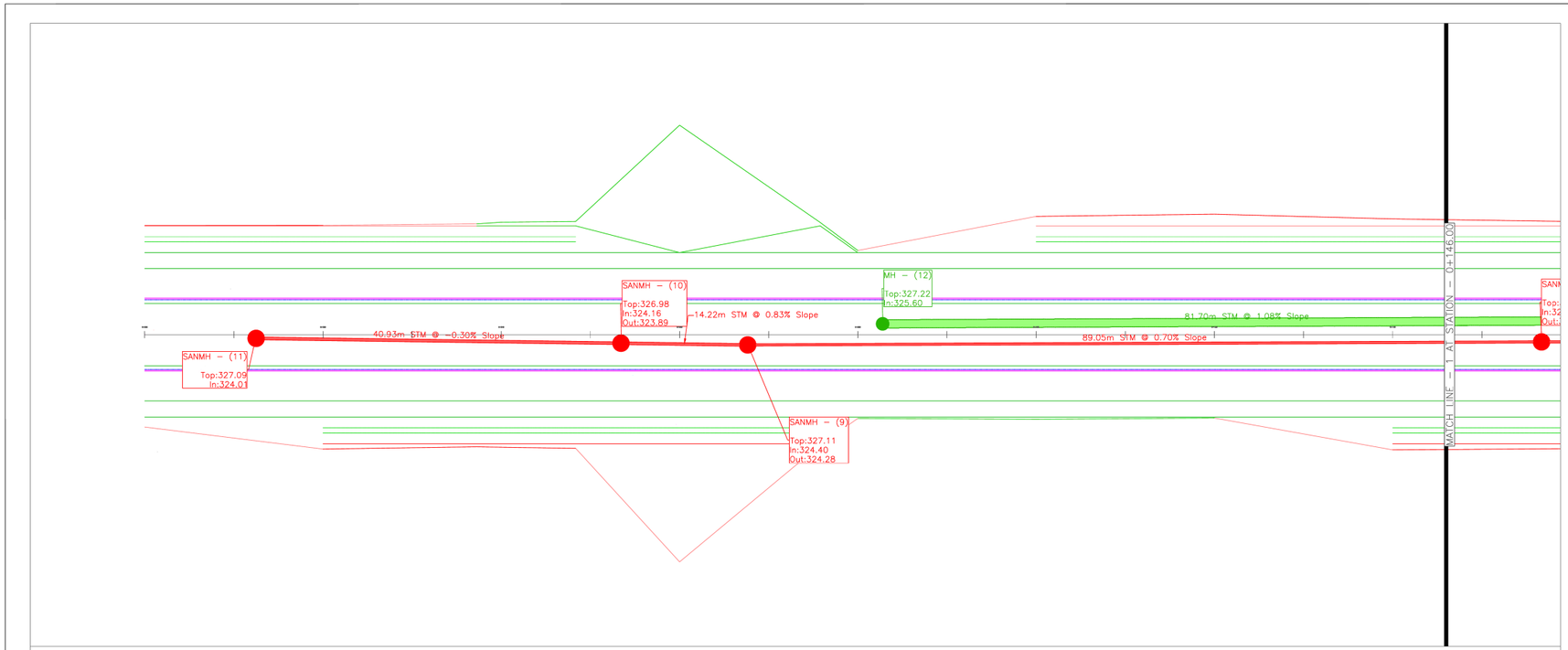
Construction to closeout

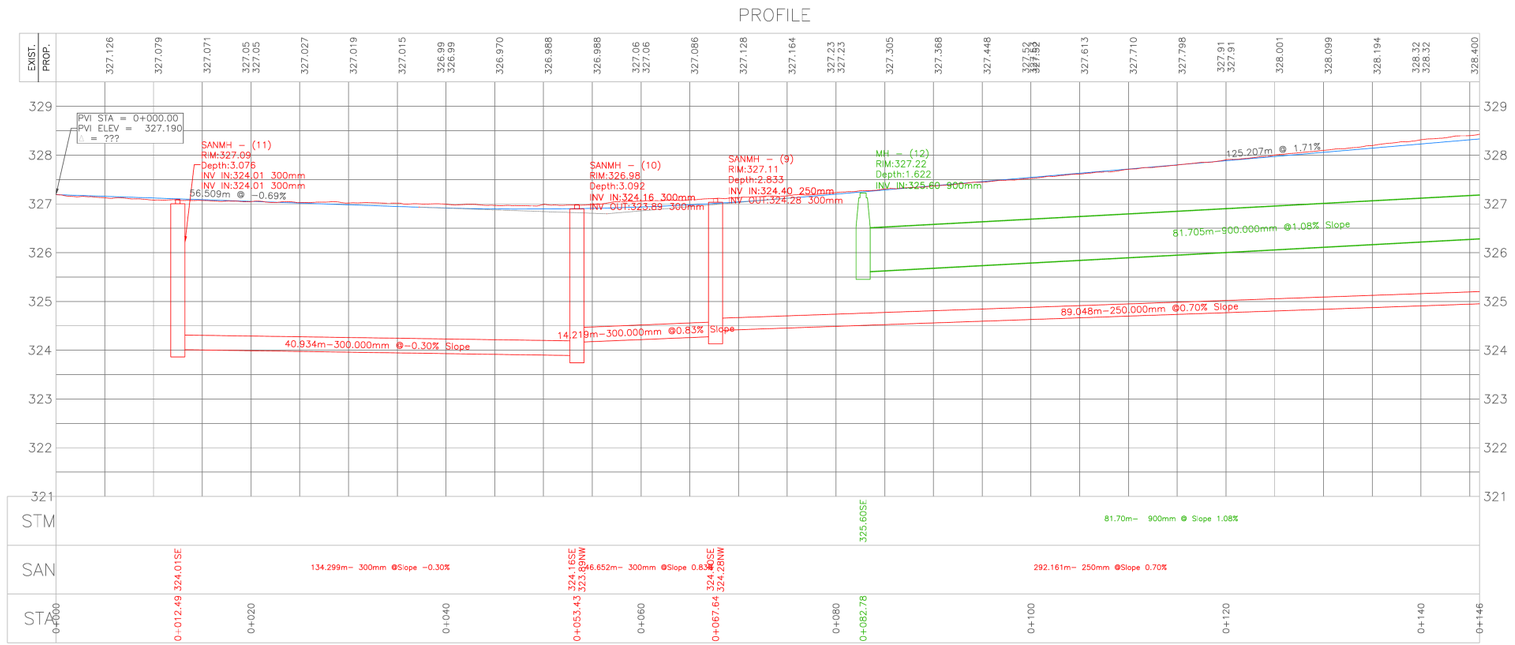
Complete inspection and testing, deficiency correction, CCTV/mandrel/leakage records as applicable, as-built information, O&M; documentation, post-construction verification and required owner/Director notifications.

PARALLEL SCREENING

Confirm applicable planning and Environmental Assessment Act obligations, conservation authority and source-protection requirements, utility and road permits, heritage/archaeology, property/easements, traffic staging, erosion and sediment control, and stakeholder consultation.

Regulatory source basis: Ontario MECP municipal CLI-ECA guidance and forms, reviewed September 2026.





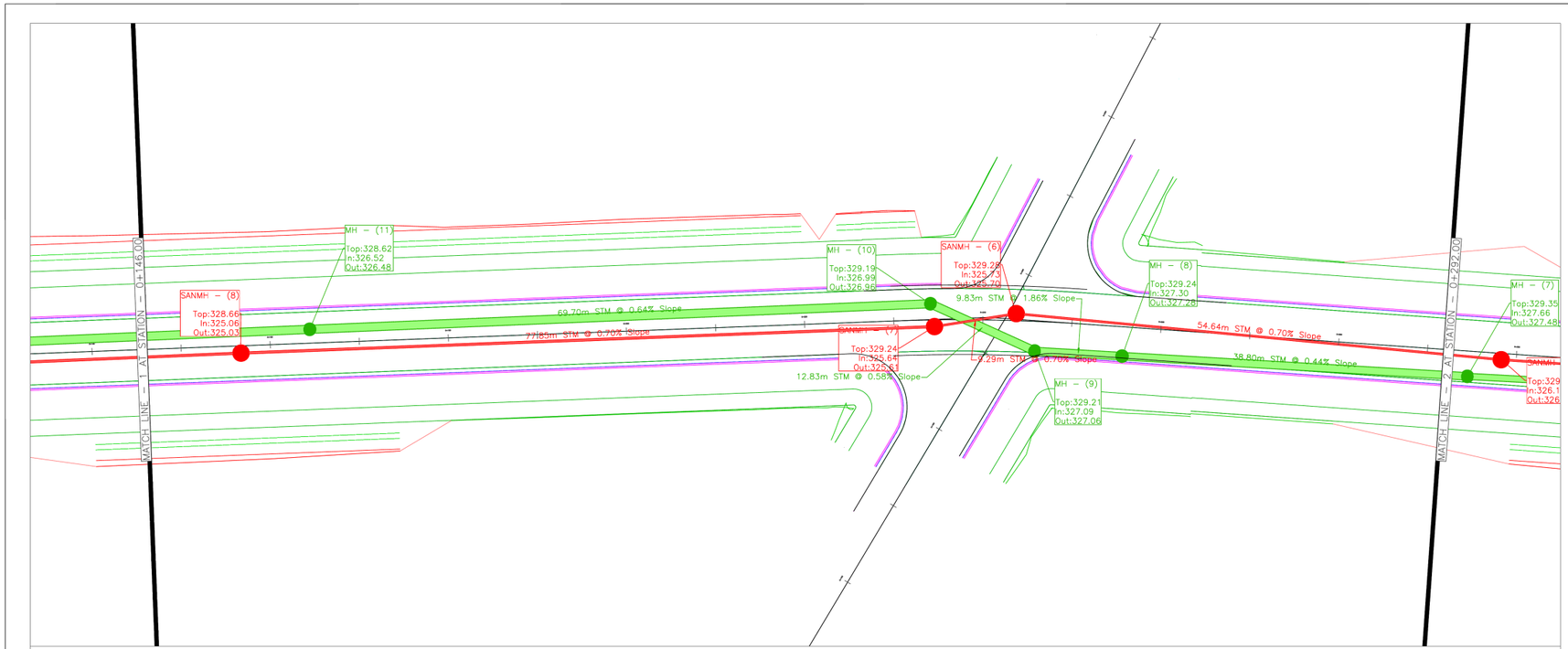
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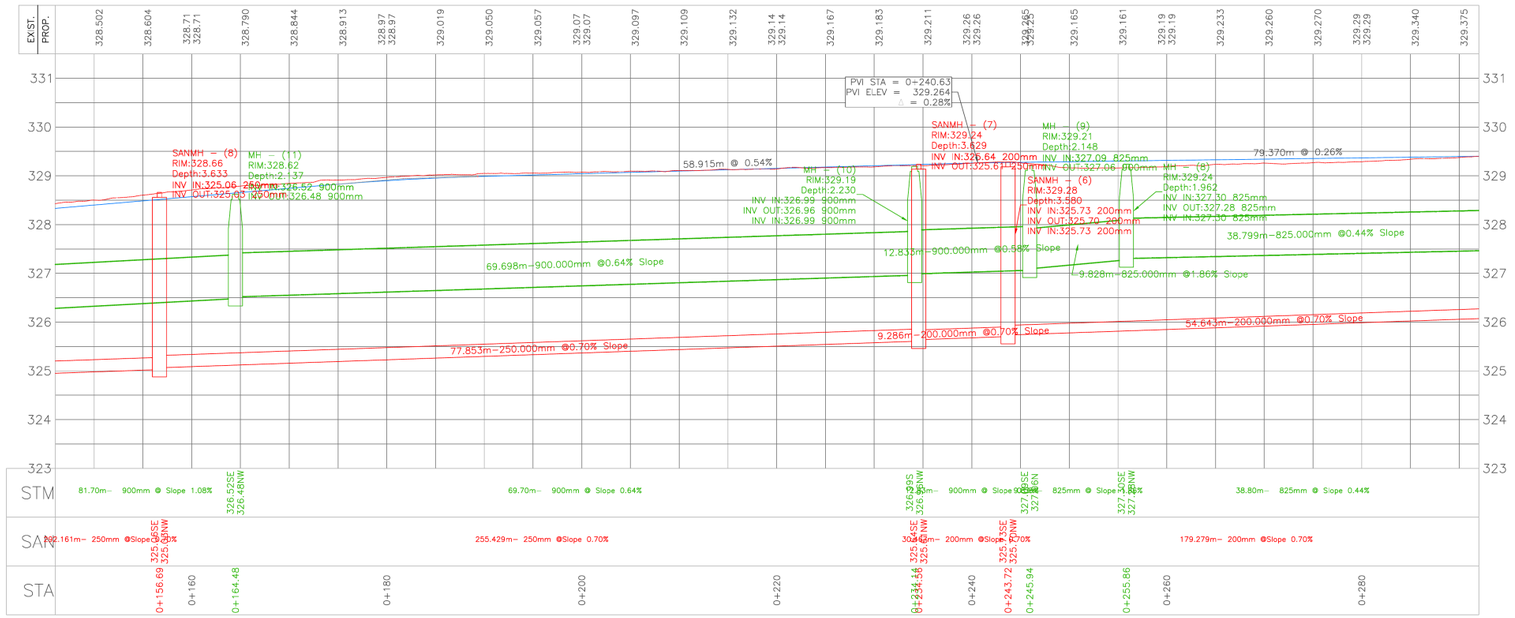
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PROFILE



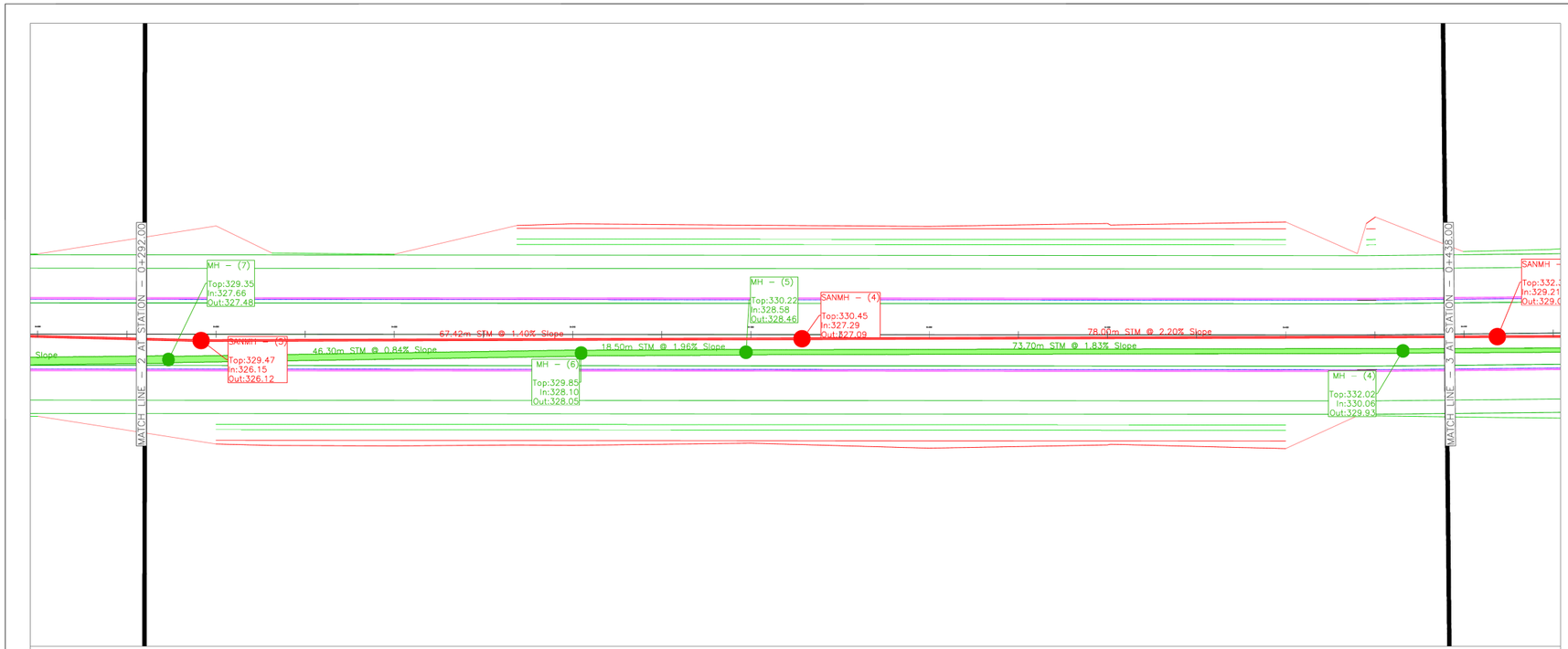
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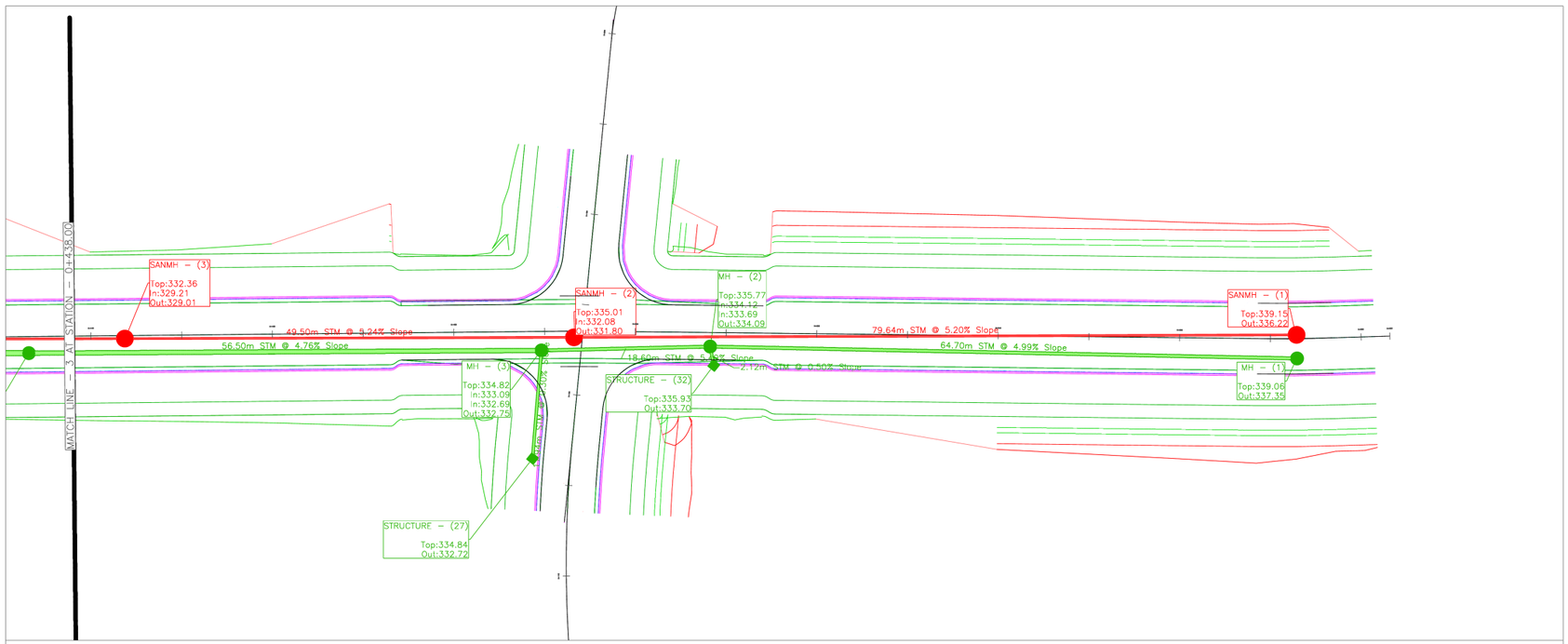
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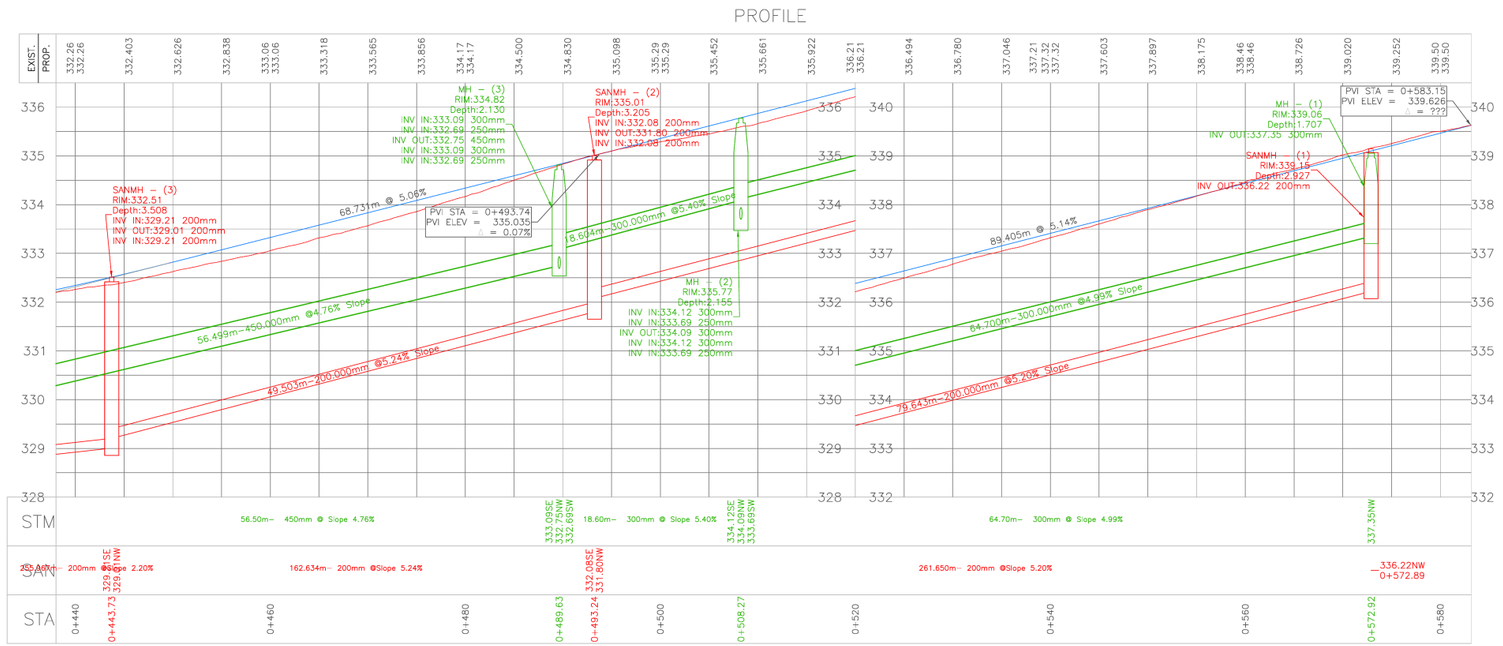
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Corridor grading and section-view production

Section views expose how the corridor behaves between control points

The section package samples the conceptual road corridor at regular stations and presents existing terrain with proposed corridor geometry. It supports review of crossfall, width transitions, intersection tie-ins and changes in roadside grading that are difficult to assess from plan and profile alone.

29

SECTION VIEWS

Stations 0+020 through 0+580

20 m

REGULAR INTERVAL

Additional critical stations should be added where required

+/-20 m

OFFSET RANGE

Approximate display range from road centreline

3

DRAWING SHEETS

Multiple-section layout for sequential review

WHAT EACH SECTION COMMUNICATES

Existing ground

Interpreted terrain across the sampled offset range.

Proposed geometry

Corridor crown, lane, gutter/curb and roadside links where represented by the assembly.

Levels and slopes

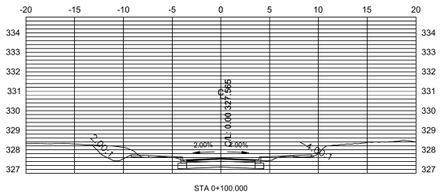
Centreline elevations and crossfall labels used to review drainage intent.

Transitions

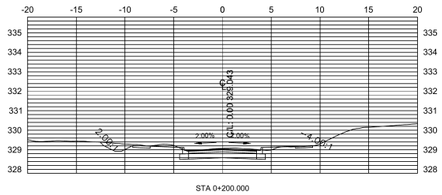
Changes in width, slope and corridor geometry near intersections and tie-ins.

DESIGN REVIEW AND LIMITS

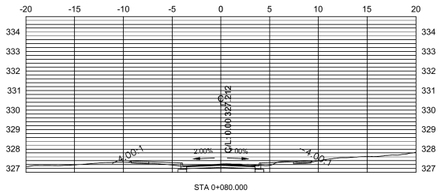
Use the sequence to check crown-to-gutter drainage, lane crossfall, assembly consistency, daylight behaviour and local transitions. These views demonstrate Civil 3D production; they do not establish final earthwork, property impacts, accessibility, utility clearances or construction-ready grading. Generic title-block fields remain placeholders.



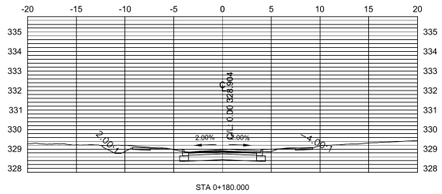
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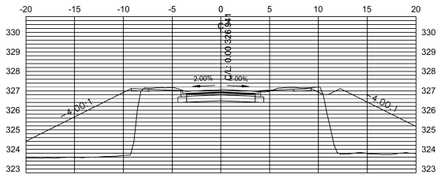
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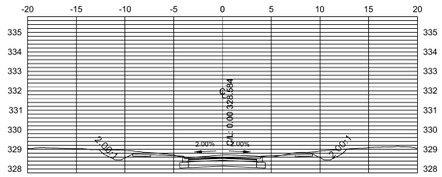
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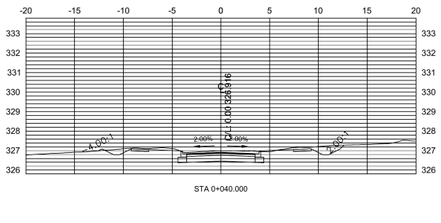
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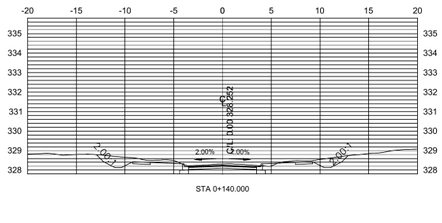
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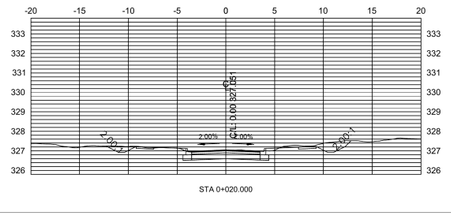
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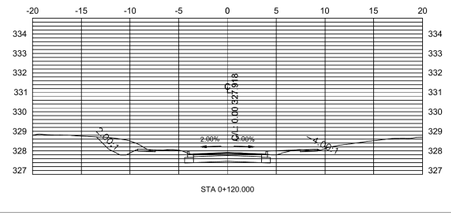
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STA 0+020.000



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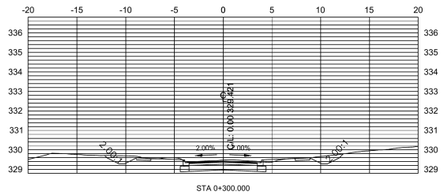
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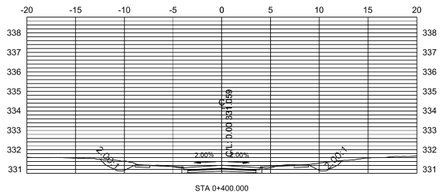
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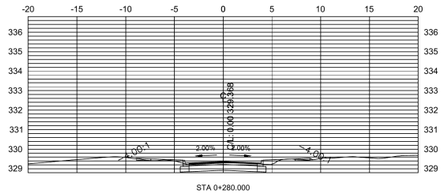
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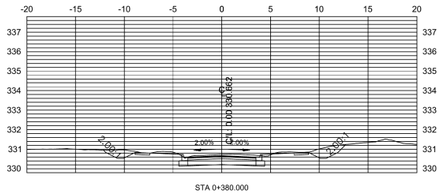
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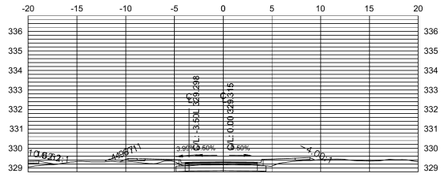
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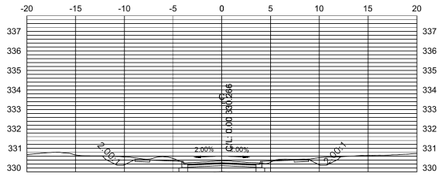
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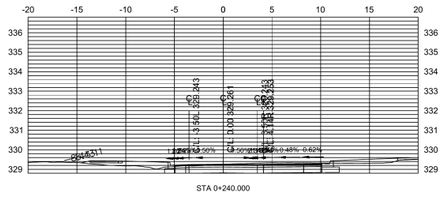
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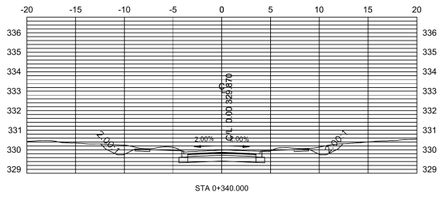
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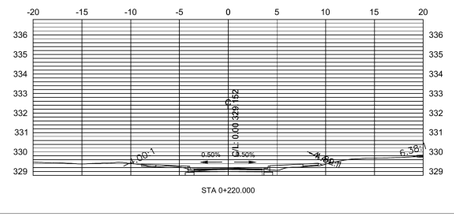
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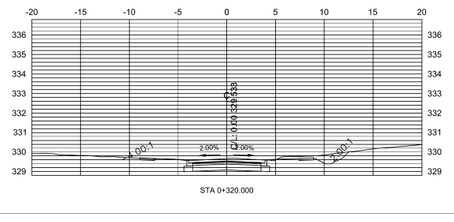
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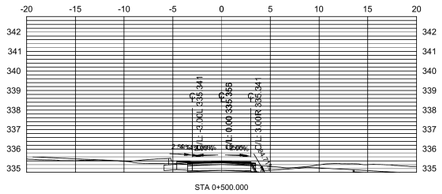
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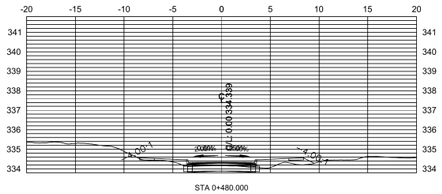
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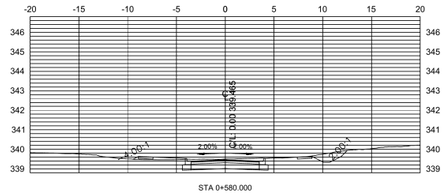
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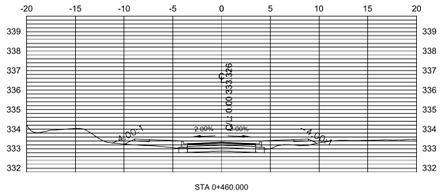
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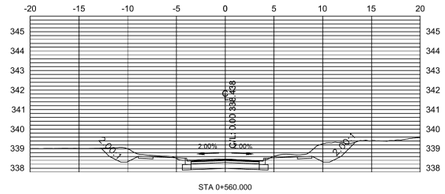
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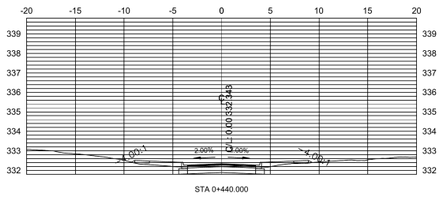


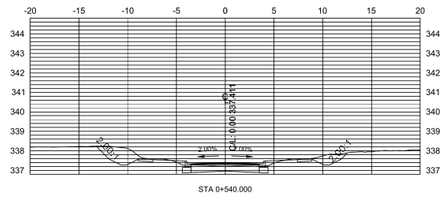


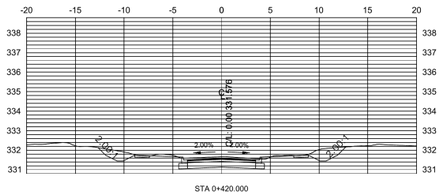


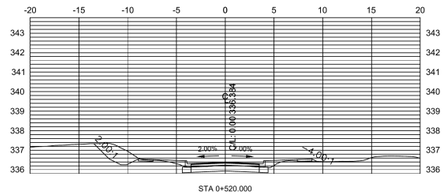












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Demonstrated work and remaining verification

A credible portfolio separates completed work from professional sign-off

The model and sheets demonstrate a substantial Civil 3D production workflow. The items at right remain mandatory before similar work could support a real municipal submission or construction package.

DEMONSTRATED IN THIS PORTFOLIO

Model organization

Existing ground, alignments, profiles, corridor and referenced design objects.

Road coordination

Intersection geometry, lane-width transitions, crown-to-gutter grading intent and tie-ins.

Network production

Storm and sanitary pipes, structures, plan/profile display, labels and bands.

Sheet delivery

Plan/profile and cross-section sheets, XREF coordination, colour plotting and publication.

REQUIRED BEFORE REAL-WORLD USE

Verified base information

Legal survey, subsurface utilities, records review, geotechnical information and field checks.

Engineering analysis

Hydrology, hydraulics, downstream reserve capacity, I&I;, pipe sizing and design exceptions.

Independent QA/QC

Object connectivity, cover, crossings, constructability, quantities, specifications and checking.

Authorization and closeout

Owner acceptance, applicable approvals, licensed review, inspection, testing and as-builts.

PORTFOLIO USE

Prepared to demonstrate Civil 3D modelling, municipal servicing coordination and drawing-production skills. It is not an issued design, permit submission, tender package, record drawing or construction document.